A COMPREHENSIVE REPORT OF MY INTERNSHIP TENURE

From 27-05-2024 to 27-07-2024

CALIBRATION OF LC DEVICES

We have conducted experiments and tests to Calibrate the LC (SCD 30) Devices to HC(Testo, Licor). To minimize the uncertainty gap between them, data analysis will be performed to study the flow of Carbon dioxide, temperature and Relative humidity in different environments, and data visualization will be performed to better understand the data.

Arham Aqeel
QuEST LAB

Contents

INTRODUCTION....3
Why did the energy crisis happen in the 1970s? 3
The effect of the 1970s energy crisis on Indoor spaces. 3
Air Contaminants sources and consequences....4
Biological Contaminants 4
Chemical Contaminants 4
Off-Gassing Emissions....4
Particulate Matter 5
Carcinogens 5
IAQ Standards. 5
Biological Contaminants 5
Chemical Contaminants and VOCs 5
Particulate Matter 6
Radon....6
IAQ Assessment Methods 6
Is CO_2 A Pollutant?....6
Ventilation and Its measuring metrics. 7
Ventilation Metrics....7
Ventilation Guidelines....7
Determining Adequate Ventilation 8
Relation of CO_2 and airborne transmission. 8
Assumptions were made for a well-mixed room....9
Significance of measuring CO_2 in both Space and Time 9
How Does it relate to my field?....11
Conclusion 12
Mathematics 13
Ideal Gas Equation in Terms of CO_2 Concentration 13
REFERENCES....15

INTRODUCTION

Why did the energy crisis happen in the 1970s?

The 1970s energy crisis occurred due to a combination of factors:

1. **Petroleum production in the U.S. and other parts of the world peaked in the late 1960s and early 1970s**, leading to genuine concerns about supply.
2. **In 1973, the Yom Kippur War interrupted Middle Eastern oil exports** when Arab OPEC members imposed an embargo on oil shipments to the U.S. and its allies.
3. **The 1979 energy crisis was triggered by the Iranian Revolution**, which caused another disruption in oil exports.
4. **The U.S. and other Western countries relied heavily on Middle Eastern oil**, making them vulnerable to supply shocks.
5. **The crisis led to stagnant economic growth and high inflation** in many countries as oil prices surged, coining the term "stagflation."
6. **The crisis prompted a shift towards energy-saving technologies and a search for alternative energy sources**, but momentum stalled in the 1980s as prices stabilized and dropped.

The energy crisis was driven by peaking oil production, supply disruptions from geopolitical events, and the West's heavy reliance on Middle Eastern oil. It had significant economic and political consequences that reverberated through the decade.

The effect of the 1970s energy crisis on Indoor spaces.

The energy crisis of the 1970s had significant implications for indoor air quality (IAQ) in buildings, driven primarily by the push for energy efficiency in response to rising energy costs. Here are the critical effects:

1. **Increased Energy Efficiency Measures**: To reduce energy consumption, buildings were retrofitted to be more energy-efficient. This often involved sealing buildings tightly to minimize air leaks, which inadvertently reduced natural ventilation. While this helped conserve energy, it also trapped indoor pollutants, leading to poorer air quality.
2. **Ventilation Changes**: The focus on energy savings reduced the amount of outdoor air introduced into buildings. This change meant that indoor air pollutants from sources like cleaning products, building materials, and occupant activities accumulated, exacerbating IAQ issues.
3. **Health Impacts**: Poor indoor air quality became a growing concern as occupants were exposed to higher concentrations of pollutants. Studies indicated that inadequate ventilation contributed to health problems, including respiratory issues and decreased productivity. The increased awareness of these health impacts led to more research and regulations regarding IAQ in the following decades.

4. **Regulatory Responses**: The energy crisis prompted the establishment of the Department of Energy and spurred legislative actions to improve energy efficiency and air quality standards. This included revising the Clean Air Act, which began to address indoor air quality alongside outdoor pollution.

Overall, the energy crisis highlighted the need for a balanced approach to energy efficiency and indoor air quality, leading to ongoing discussions and developments in building design and ventilation strategies.

Air Contaminants sources and consequences.

Indoor air contaminants can significantly impact health and well-being. Here's a breakdown of various types of pollutants, their sources, and potential consequences:

Biological Contaminants

Biological contaminants, or bio-contaminants, include microorganisms such as bacteria, fungi, viruses, and allergens from pets and insects. Common sources include:

- **Mold spores** from damp areas
- **Bacteria** from surfaces and air
- **Pollen** and **animal dander**

These contaminants can lead to allergic reactions, respiratory issues, and infections. For example, mold can exacerbate asthma and other respiratory conditions, while bacteria and viruses can spread diseases.

Chemical Contaminants

Chemical contaminants comprise volatile organic compounds (VOCs) and other gases released from various materials. Common sources include:

- **Household products** like paints, cleaners, and air fresheners
- **Building materials** such as pressed wood and carpets

VOCs can cause headaches, dizziness, and long-term health issues, including cancer. For instance, formaldehyde, a standard VOC, is a known carcinogen and can lead to respiratory problems and irritation of the eyes and throat.

Off-Gassing Emissions

Off-gassing refers to releasing gases from solid or liquid materials, particularly new furniture, carpets, and construction materials.

These emissions can contribute to indoor air pollution, leading to respiratory problems, headaches, and long-term health risks. Prolonged exposure can result in chronic health issues, particularly in poorly ventilated spaces.

Particulate Matter

Particulate matter includes tiny particles suspended in the air, which can originate from various sources, including:

- **Dust** from indoor activities
- **Smoke** from cooking or heating
- **Outdoor pollutants** that enter buildings

Exposure to particulate matter can cause respiratory diseases, cardiovascular issues, and asthma exacerbation. Fine particles can penetrate the lungs and even enter the bloodstream, leading to serious health problems.

Carcinogens

Carcinogens are substances that can lead to cancer, and several indoor air contaminants fall into this category, including:

- **Formaldehyde** from building materials
- **Radon**, a naturally occurring gas that can seep into homes from the ground
- **Asbestos** fibers from older building materials

Exposure to carcinogens can increase the risk of developing various cancers, including lung cancer, from radon exposure. Asbestos exposure is linked to mesothelioma and other serious lung diseases. In summary, indoor air contaminants can arise from biological, chemical, and particulate sources, each posing unique health risks. Adequate ventilation and using low-emission materials can help mitigate these risks and improve indoor air quality.

IAQ Standards.

Several essential IAQ standards and guidelines address various indoor air contaminants:

Biological Contaminants

- **WHO guidelines** provide thresholds for biological contaminants like mold and bacteria.
- **ASHRAE Standard 62.1** includes requirements for ventilation to dilute and remove indoor pollutants, including biological contaminants.

AIR QUALITY INDEX

	AQI	PM 2.5 (ug/m3)	PM 10 (ug/m3)	VOC (ppm)	CO2 (ppm)	Formaldehyde (ppm)
Good	0-50	0 - 12	0 - 54	0 - 15	400 - 650	0 - 0.2
Moderate	51 - 100	12.1 - 35.4	55 - 154	16 - 25	651 - 1500	0.21 - 0.4
Unhealthy for sensitive groups	101 - 150	35.5 - 55.4	155 - 254	26 - 50	1501 - 2000	0.41 - 0.6
Unhealthy	151 - 200	55.5 - 150.4	255 - 354	51 - 75	2001 - 2500	0.61 - 0.8
Very Unhealthy	201 - 300	150.5 - 250.4	355 - 424	76 - 100	2501 - 5000	0.81 - 1
Hazardous	301 - 500	250.5 - 500	425 - 600	101 - 150	5001 - 15000	1.01 - 1.2

Chemical Contaminants and VOCs

- **WHO guidelines** set limits for VOCs like benzene and formaldehyde.

- **ASHRAE Standard 62.1** specifies maximum CO_2levels of 1000 ppm to indicate adequate ventilation to control VOCs and other contaminants.
- **Germany and the Netherlands have formaldehyde guidelines** of 120 µg/m^3 for continuous exposure.

Particulate Matter

- **WHO guidelines** provide thresholds for particulate matter (PM2.5 and PM10).
- **ASHRAE Standard 62.1** requires minimum efficiency reporting value (MERV) ratings for air filters to remove particles.

Radon

- **WHO recommends an action level of 100 Bq/m^3** for radon, with a target of 300 Bq/m^3.
- **Canada's Atomic Energy Control Board has criteria for radon cleanup** with a target of 0.02 WL (working level) or 0.01 WL for the annual average.

IAQ Assessment Methods

- **Measurement of CO_2levels** is commonly used as a proxy for ventilation adequacy and occupant comfort.
- **Sampling and analysis of air for specific contaminants** like VOCs, particulates, and biological agents can assess compliance with IAQ standards.
- **Occupancy surveys and symptom logs** can help identify IAQ problems and their health impacts.

While OSHA does not have specific IAQ standards, the General Duty Clause requires employers to provide a safe workplace free of known hazards. Adhering to consensus standards like ASHRAE 62.1 and WHO guidelines help ensure IAQ meets health and safety requirements.

Is CO_2 A Pollutant?

Yes, carbon dioxide (CO_2) is considered a pollutant in indoor environments, primarily due to its potential health impacts at elevated levels.

Why $CO_2$is Considered a Pollutant

1. **Health Effects**: While CO_2 is a naturally occurring gas essential for life, high concentrations can lead to health issues. At levels above 1000 ppm, CO_2can cause symptoms such as headaches, dizziness, shortness of breath, and impaired cognitive function. Prolonged exposure to even higher levels can lead to more severe health effects, including respiratory problems and unconsciousness.
2. **Indicator of Ventilation**: $CO_2$is often used to indicate indoor air quality and ventilation effectiveness. High levels of CO_2typically suggest inadequate ventilation, which can allow other indoor pollutants to accumulate, further compromising air quality and occupant health.

Sources of CO_2 in Indoor Spaces

CO_2 is produced from various sources in indoor environments, including:

- **Human Respiration**: Everyone exhales CO_2, making it a significant source in occupied spaces.
- **Combustion Appliances**: Gas stoves, heaters, and fireplaces release CO_2 during fuel combustion. Poorly vented appliances can significantly increase indoor CO_2 levels.
- **Building Materials**: Some materials can release CO_2 as they age or degrade, although this is typically a minor source compared to respiration and combustion.
- **Occupant Activities**: Cooking and burning candles also contribute to indoor CO_2 levels.

In summary, while CO_2 is a natural component of the atmosphere, it becomes a pollutant in indoor spaces when concentration rises due to insufficient ventilation and other sources, leading to potential health risks for occupants.

Ventilation and Its measuring metrics.

There are several key metrics and guidelines for measuring and determining adequate ventilation rates in buildings to mitigate airborne disease transmission:

Ventilation Metrics

1. **Air Changes per Hour (ACH)**: The number of times the air volume of a space is replaced per hour. Typical recommendations range from 4-6 ACH for general spaces to 15-60 ACH for kitchens and other high-pollution areas.
2. **Ventilation Rate per Person**: The volume of outdoor air supplied per person per unit time, often expressed as liters per second per person (L/s-person) or cubic feet per minute per person (CFM/person). For example, ISO 17772 specifies 10 L/s-person for high indoor air quality in residences.
3. **Ventilation Rate per Area**: The volume of outdoor air supplied per unit floor area per unit time, such as L/s-m2 or CFM/ft2. This metric is helpful for spaces with variable occupancy.

Ventilation Guidelines

1. **CSIR India**: Provides equations and tables to estimate minimum ventilation rates for natural and mechanical systems based on wind speed and room dimensions.
2. **ASHRAE**: Recommends a minimum of 4-6 ACH for most spaces, with higher rates for high-occupancy or high-pollution areas. ASHRAE also specifies minimum ventilation rates per person.
3. **CDC**: Advises increasing ventilation rates and filtration efficiency to the extent possible. Suggests getting at least 5 ACH of clean air in occupied spaces.
4. **REHVA**: Recommends increasing ventilation rates, using demand-controlled ventilation, and improving filtration to MERV 14 or higher.

5. **US EPA**: Suggests increasing ventilation rates, improving filtration, and using portable air cleaners to supplement ventilation.

Determining Adequate Ventilation

CDC, ASHRAE, and other organizations recommend aiming for at least 5-6 ACH of clean air in occupied spaces, with higher rates for high-risk situations. Factors to consider include:

- **Occupancy levels and activity levels**
- **Presence of high-emission sources**
- **Filtration efficiency and air cleaning**
- **Outdoor air quality**
- **Climate and energy costs**

Ultimately, the goal is to dilute and remove airborne contaminants to acceptable levels while balancing other factors. Monitoring CO_2levels can help assess ventilation adequacy, as higher CO_2indicates poor ventilation and higher airborne disease transmission risk.

Relation of CO_2 and airborne transmission.

CO_2concentrations and airborne transmission of diseases are closely related, particularly in indoor environments where ventilation and air quality play critical rolecs in the spread of infectious agents.

Relationship Between CO_2Concentrations and Airborne Transmission

1. **Indicator of Ventilation**: Elevated CO_2levels in indoor spaces often indicate poor ventilation. When CO_2concentrations rise, it suggests that the air is not being adequately exchanged, allowing airborne pathogens to accumulate. Studies have shown that higher CO_2levels correlate with increased concentrations of airborne pathogens, thereby elevating the risk of transmission of respiratory diseases such as COVID-19.

2. **Exposure Time and Infectious Dose**: The risk of airborne transmission is influenced by the concentration of pathogens in the air and the duration of exposure. CO_2monitoring can help assess the risk by providing real-time data on air quality. For instance, a study developed a model that connects CO_2levels to the probability of airborne transmission of SARS-CoV-2, indicating that maintaining CO_2concentrations below certain thresholds can help mitigate transmission risk.

3. **Guidelines for Safe Levels**: Research suggests specific CO_2concentration limits to minimize transmission risks. For example, in clinical settings, maintaining CO_2levels below 620 ppm without masks and below 1000 ppm with surgical masks can help reduce the primary reproduction number (R0) of airborne diseases.

4. **Environmental Conditions**: The physical environment, including humidity and temperature, can affect both CO_2levels and the viability of airborne pathogens. Poor

ventilation not only raises CO_2 but also allows pathogens to linger longer in the air, increasing the likelihood of infection among susceptible individuals.

In summary, monitoring CO_2 concentrations is a practical method for assessing indoor air quality and the associated risk of airborne disease transmission. Ensuring adequate ventilation and maintaining CO_2 levels within safe limits can significantly reduce airborne transmission risk, thus enhancing indoor health and safety.

Assumptions were made for a well-mixed room.

The well-mixed room or well-mixed assumption is fundamental in modeling airborne disease transmission in indoor spaces. It assumes that the concentration of airborne pathogens is uniform throughout the room, with no spatial variations. This assumption simplifies the analysis and allows for the use of well-established models from chemical engineering to predict the spread of infectious diseases. The well-mixed assumption is essential for several reasons:

1. **It provides a baseline for understanding airborne transmission dynamics** in indoor spaces.
2. **Many existing models and guidelines for mitigating airborne transmission rely on this assumption**.
3. **It allows for developing simple, practical guidelines** for limiting indoor airborne transmission, such as minimum ventilation rates and occupancy limits.

However, the validity of the well-mixed assumption in real-world indoor spaces has been questioned. Some studies suggest that large rooms may not achieve a well-mixed state, leading to spatial variations in pathogen concentrations. This is because the mixing time, which depends on the room size and ventilation characteristics, may be longer than the timescale of pathogen removal by ventilation or sedimentation.

In the context of the COVID-19 pandemic, the well-mixed assumption has been used to develop guidelines for limiting indoor airborne transmission. However, these guidelines may need to be adjusted based on the specific characteristics of each room, such as size, shape, and ventilation system.

While the well-mixed assumption provides a helpful starting point for understanding airborne transmission, it is essential to recognize its limitations, especially in large rooms or spaces with complex ventilation systems. More detailed modeling and experimental studies are needed to understand better the spatial distribution of airborne pathogens in real-world indoor environments and to develop more accurate guidelines for mitigating airborne transmission risks.

Significance of measuring CO_2 in both Space and Time

Monitoring carbon dioxide (CO_2) levels in indoor spaces is significant for several reasons, particularly air quality and health outcomes. Here's an overview of its importance, the implications of monitoring over time and space, and its relationship to airborne transmission of diseases.

Significance of Monitoring CO_2

1. **Indicator of Ventilation**: CO_2levels directly indicate indoor air quality and ventilation effectiveness. High CO_2concentrations often indicate inadequate ventilation, which can lead to the accumulation of other harmful pollutants, including airborne pathogens.
2. **Health Risks**: Elevated CO_2levels can cause various health issues, including headaches, dizziness, fatigue, and impaired cognitive function. Monitoring allows timely interventions to improve air quality and reduce health risks associated with high CO_2concentrations.
3. **Airborne Disease Transmission**: Monitoring CO_2can help assess the risk of airborne transmission of diseases such as COVID-19. Studies indicate that higher CO_2levels correlate with increased concentrations of airborne pathogens, making it essential to maintain adequate ventilation to mitigate these risks.
4. **Energy Efficiency**: CO_2monitoring can optimize ventilation systems in energy-efficient buildings. By adjusting ventilation rates based on real-time CO_2levels, buildings can maintain healthy air quality while minimizing energy consumption.

Monitoring in Space and Time

1. **Spatial Monitoring**: Assessing CO_2levels across different building areas can help identify poorly ventilated zones. This spatial data is crucial for implementing targeted ventilation strategies, particularly in larger or multi-room environments where air mixing may not be uniform.
2. **Temporal Monitoring**: Continual monitoring provides insights into how CO_2levels fluctuate with occupancy patterns, activities, and ventilation system performance. This temporal data can inform adjustments to ventilation strategies and help maintain optimal indoor air quality throughout varying conditions.

Achieving Well-Mixed Conditions

While the well-mixed assumption simplifies modeling airborne transmission, real-world conditions often refute this assumption, especially in large spaces. In such environments, CO_2levels may not be uniform due to factors like:

- **Room Size and Shape**: Larger rooms may have areas with stagnant air where CO_2can accumulate, leading to spatial variations in concentration.
- **Ventilation Design**: Inefficient or poorly designed ventilation systems can create dead zones where air does not circulate effectively.

Conclusion

In conclusion, monitoring CO_2concentrations in space and time is crucial for maintaining good indoor air quality, reducing health risks, and mitigating airborne disease transmission. While the well-mixed assumption provides a valuable framework, it is essential to consider the unique characteristics of each indoor space to ensure adequate ventilation and air quality management. Continuous CO_2monitoring can guide interventions to optimize indoor environments' health outcomes and energy efficiency.

How Does it relate to my field?

The topics discussed in the previous questions are intricately related to mechanical engineering,

How these elements connect to mechanical engineering and their future relevance:

Connection to Mechanical Engineering

1. **Ventilation Systems Design**: Mechanical engineers are responsible for designing and optimizing building ventilation systems. This includes calculating appropriate air changes per hour (ACH) and ensuring systems effectively manage CO_2levels and other indoor pollutants. Understanding the principles of fluid dynamics and thermodynamics is crucial in this context.

2. **Air Quality Management**: Engineers develop technologies and systems to monitor and improve indoor air quality. This involves integrating sensors for $CO_2$and other pollutants into HVAC (Heating, Ventilation, and Air Conditioning) systems and designing systems that can adapt to varying occupancy levels and activities.

3. **Thermal Comfort and Energy Efficiency**: Mechanical engineers work on balancing indoor air quality and ensuring energy efficiency. This includes designing systems that optimize energy use while providing adequate ventilation, which is increasingly crucial in sustainability and green building practices.

4. **Research and Development**: Mechanical engineers are involved in R&D to develop new materials, filtration technologies, and energy-efficient systems that improve indoor air quality. This includes exploring advanced HVAC technologies, such as demand-controlled ventilation and air purification systems.

Future Relevance

1. **Emerging Technologies**: Integrating IoT (Internet of Things) and innovative technologies in building management systems is becoming increasingly important. Mechanical engineers must adapt to these advancements, developing systems that communicate and respond to real-time air quality and occupancy data.

2. **Public Health Awareness**: The COVID-19 pandemic has heightened awareness of airborne transmission and the importance of indoor air quality. As a result, there will be greater demand for mechanical engineers who can design systems that mitigate transmission risks while ensuring comfort and energy efficiency.

3. **Sustainability Goals**: As building codes and standards evolve to prioritize sustainability, mechanical engineers will be crucial in developing energy-efficient systems that meet indoor air quality standards. This includes designing systems that utilize renewable energy sources and minimize environmental impact.

4. **Interdisciplinary Collaboration**: The future of mechanical engineering will increasingly involve collaboration with other disciplines, such as environmental engineering, architecture, and public health. Engineers must work together to create holistic approaches to building design that prioritize human health and ecological sustainability.

The relationship between indoor air quality, CO_2monitoring, and disease transmission is highly relevant to mechanical engineering. As the field evolves, mechanical engineers will be at the forefront of developing innovative solutions that address the challenges of indoor air quality, public health, and sustainability, making their expertise crucial for the future of building design and management.

Conclusion

In conclusion, maintaining good indoor air quality (IAQ) is essential for the health and well-being of individuals, particularly given that people spend a significant portion of their time indoors. Poor IAQ can lead to various health issues, including respiratory problems, allergies, and decreased productivity. Practical strategies to improve IAQ include proper ventilation, air purifiers, regular cleaning, and monitoring air quality to control pollutants.

The significance of monitoring carbon dioxide (CO_2) levels cannot be overstated, as elevated CO_2concentrations often indicate inadequate ventilation, which can lead to the accumulation of harmful airborne pathogens. Ensuring adequate ventilation rates and employing modern HVAC systems with effective filtration and purification technologies can optimize indoor environments for health and comfort.

As awareness of the impact of indoor air quality on public health continues to grow, the development and implementation of comprehensive strategies to mitigate indoor air pollution will remain a critical focus for researchers, engineers, and policymakers alike. This holistic approach will enhance individual well-being and contribute to broader public health objectives in the future.

Mathematics

Ideal Gas Equation in Terms of CO_2 Concentration

Let,

$P = total\ pressure\ of\ an\ air\ parcel\ (containing\ both\ CO_2\ and\ other\ gases),$

$V = volume\ of\ the$ chamber $(given\ as\ 1\ m^3),$

$T = absolute\ temperature\ in\ Kelvin, and$

$n_{(total)} = total\ number\ of\ moles\ in\ the\ parcel.$

The ideal gas law is:

$$P\,V = n_{(total)} R\,T \qquad (1)$$

If the CO_2 concentration (by mole) is given by

$$\chi_{(CO_2)} = \frac{n_{(CO_2)}}{n_{(total)}}$$

then the partial pressure of CO_2 is given by Dalton's law:

$$P_{(CO_2)} = \chi_{(CO_2)} \cdot \mathrm{P} \qquad (2)$$

Also, writing the ideal gas law specifically for the CO_2 component (assuming the volume is the same for each gas) we have:

$$P_{(CO_2)}\,V = n_{(CO_2)}\,R\,T \qquad (3)$$

Ideal Gas Equation in Terms of CO_2 Concentration

Chamber Specifications

Volume of the chamber:

$$V = 0.55\,\mathrm{m} \times 0.55\,\mathrm{m} \times 0.55\,\mathrm{m} = 0.166\,\mathrm{m}^3$$

Case 1: Delhi (25°C, 298.15 K)

Step 1: Estimate Barometric Pressure

Delhi's elevation 216 m. Using the simplified barometric formula:

$$P_{\mathrm{Delhi}} \approx 98.7\,\mathrm{kPa}.$$

Step 2: Partial Pressure of CO_2 (8000 ppm)

Mole fraction of CO_2:

$$\chi_{CO_2} = \frac{8000}{10^6} = 0.008.$$

Total pressure after adding CO_2:

$$P_{\mathrm{total}} = \frac{98.7\,\mathrm{kPa}}{1 - 0.008} = 99.5\,\mathrm{kPa}.$$

Partial pressure of CO_2:

$$P_{CO_2} = 0.008 \times 99.5 = 0.796\,\mathrm{kPa}.$$

Pressure difference:

$$\Delta P = 99.5\,\mathrm{kPa} - 98.7\,\mathrm{kPa} = 0.8\,\mathrm{kPa}.$$

Case 2: Drass, Jammu and Kashmir (10°C, 283.15 K)

Step 1: Estimate Barometric Pressure
Drass elevation 3280 m:

$$P_{\text{Drass}} \approx 67.5\,\text{kPa}.$$

Step 2: Total Pressure with CO_2

$$P_{\text{total}} = \frac{67.5\,\text{kPa}}{1 - 0.008} = 68.1\,\text{kPa}.$$

$$\Delta P = 68.1\,\text{kPa} - 67.5\,\text{kPa} = 0.6\,\text{kPa}.$$

Case 3: Tiruchirappalli (36°C, 309.15 K)

Step 1: Estimate Barometric Pressure
Tiruchirappalli elevation 88 m:

$$P_{\text{Tiruchi}} \approx 100.2\,\text{kPa}.$$

Step 2: Total Pressure with CO_2

$$P_{\text{total}} = \frac{100.2\,\text{kPa}}{1 - 0.008} = 101.1\,\text{kPa}.$$

$$\Delta P = 101.1\,\text{kPa} - 100.2\,\text{kPa} = 0.9\,\text{kPa}.$$

3. Uncertainty Analysis for Lens Equation

The lens equation is:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}.$$

Assume $u = v$. Then:

$$f = \frac{u}{2}.$$

Uncertainty in u propagates to f as:

$$\Delta f = \frac{1}{2}\Delta u.$$

Instrument A:

$$\Delta u = \pm\,(0.01 + 0.01u)\ \text{m}.$$

Instrument B:

$$\Delta u = \pm 0.0125u\,\text{m}.$$

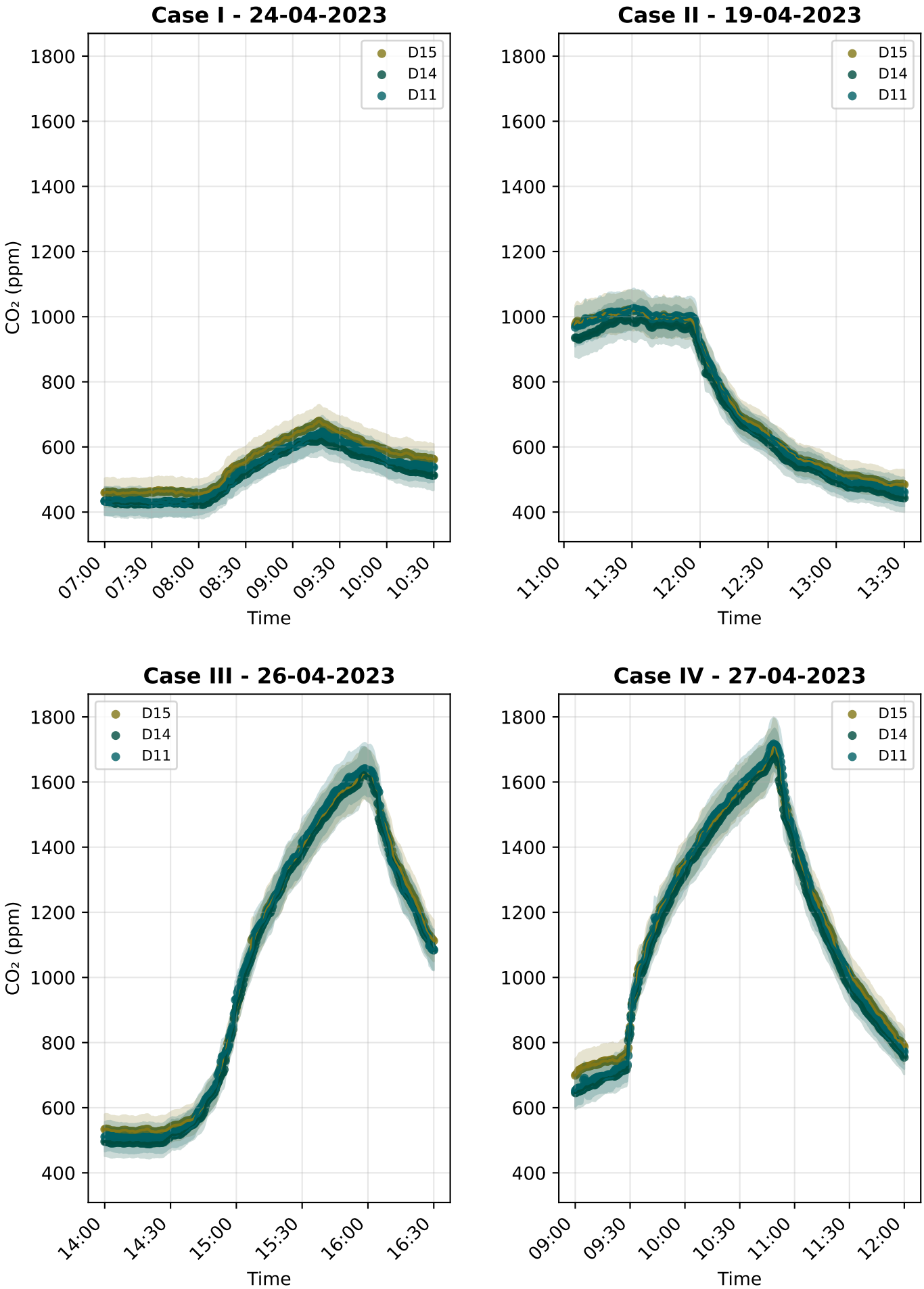
Example Calculation for $u = 1$ m:

- Instrument A: $\Delta u = 0.02\,\text{m} \Rightarrow \Delta f = 0.01\,\text{m}$.
- Instrument B: $\Delta u = 0.0125\,\text{m} \Rightarrow \Delta f = 0.00625\,\text{m}$.

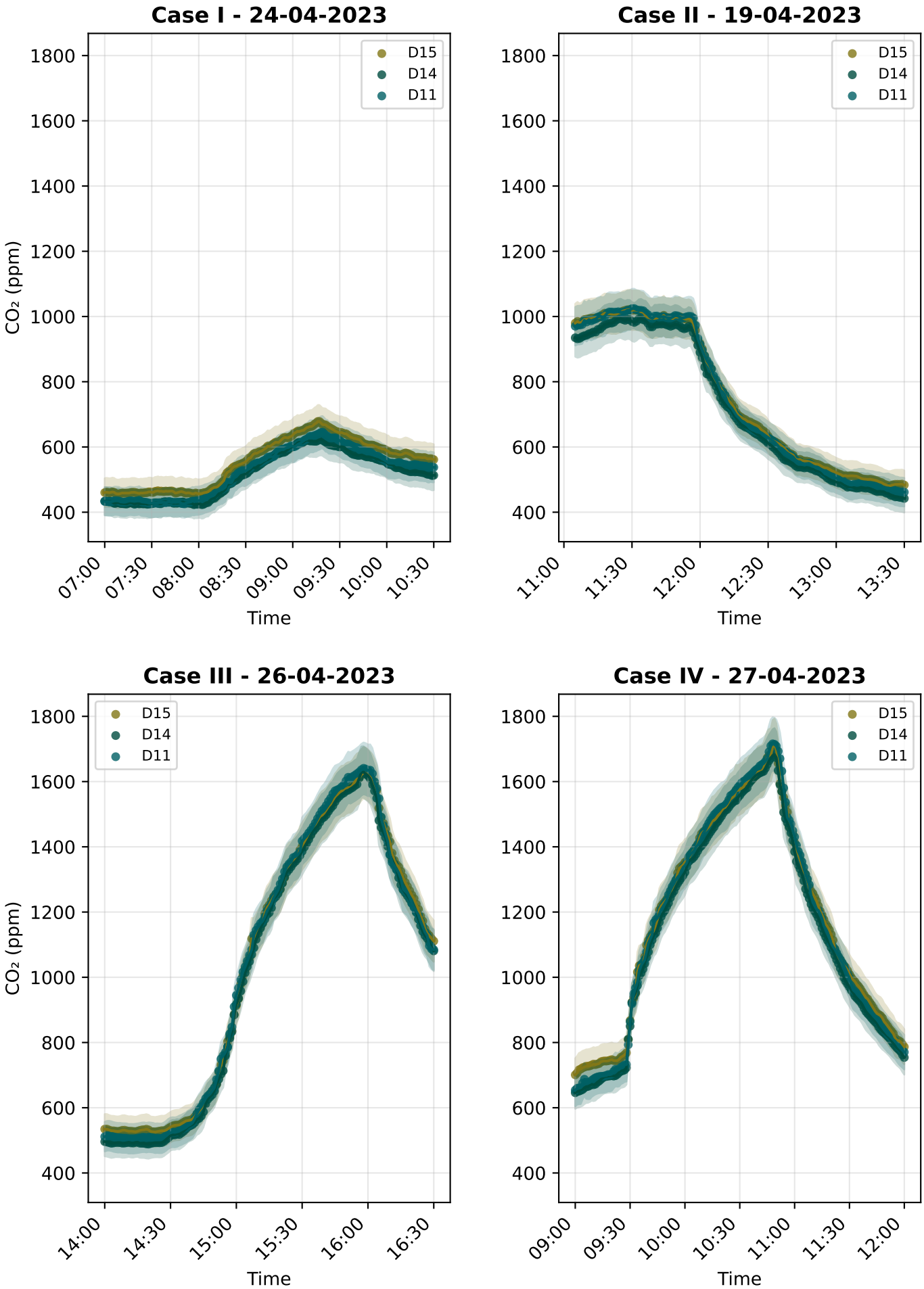
Conclusion: Instrument B provides lower uncertainty. Use **Instrument B**.

CO_2 (ppm) Time Series
Sensors 15, 14, and 11

CO_2 (ppm) - 30-Second Resolution - Sensors 15, 14, and 11

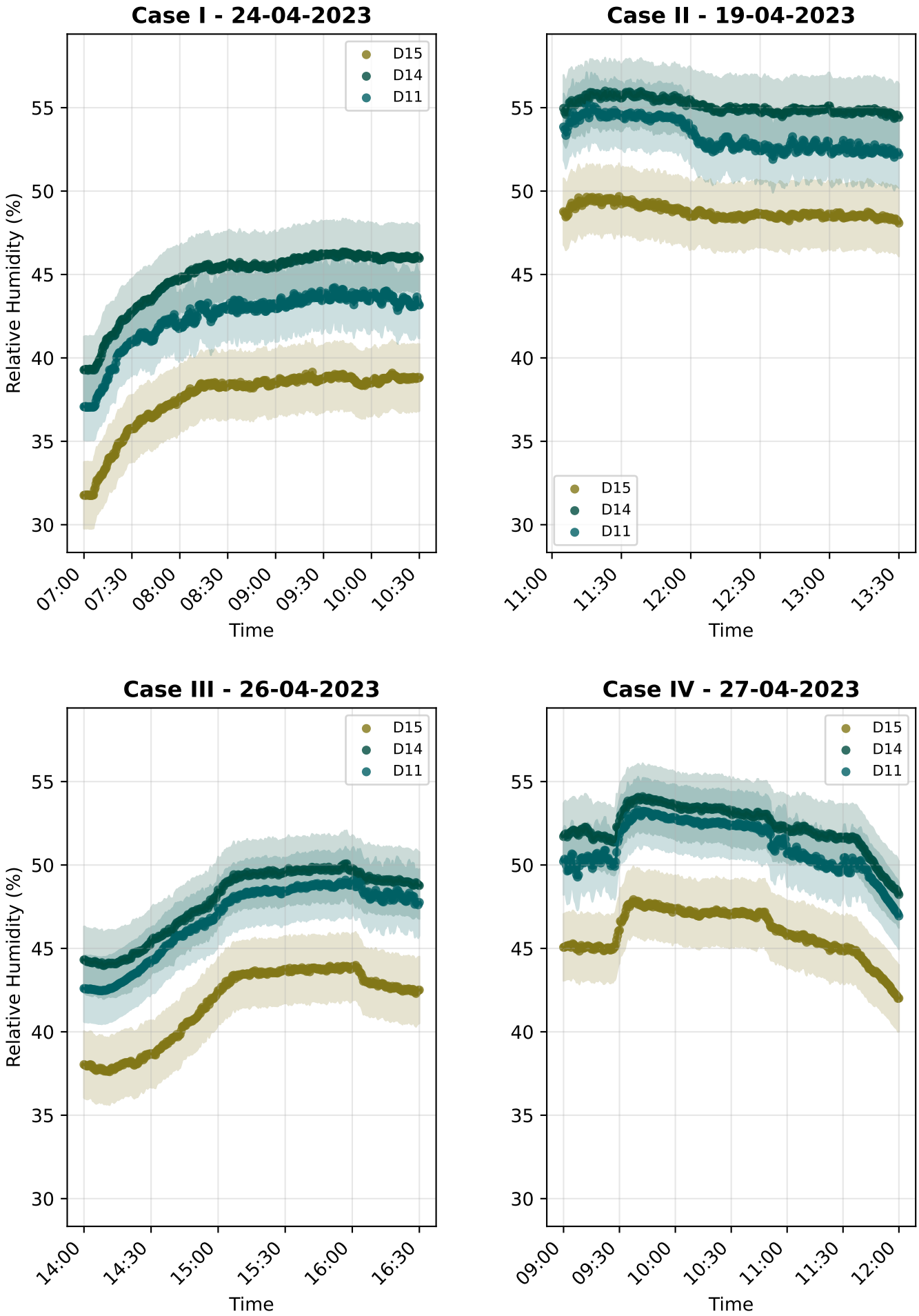


CO_2 (ppm) - 1-Minute Resolution - Sensors 15, 14, and 11

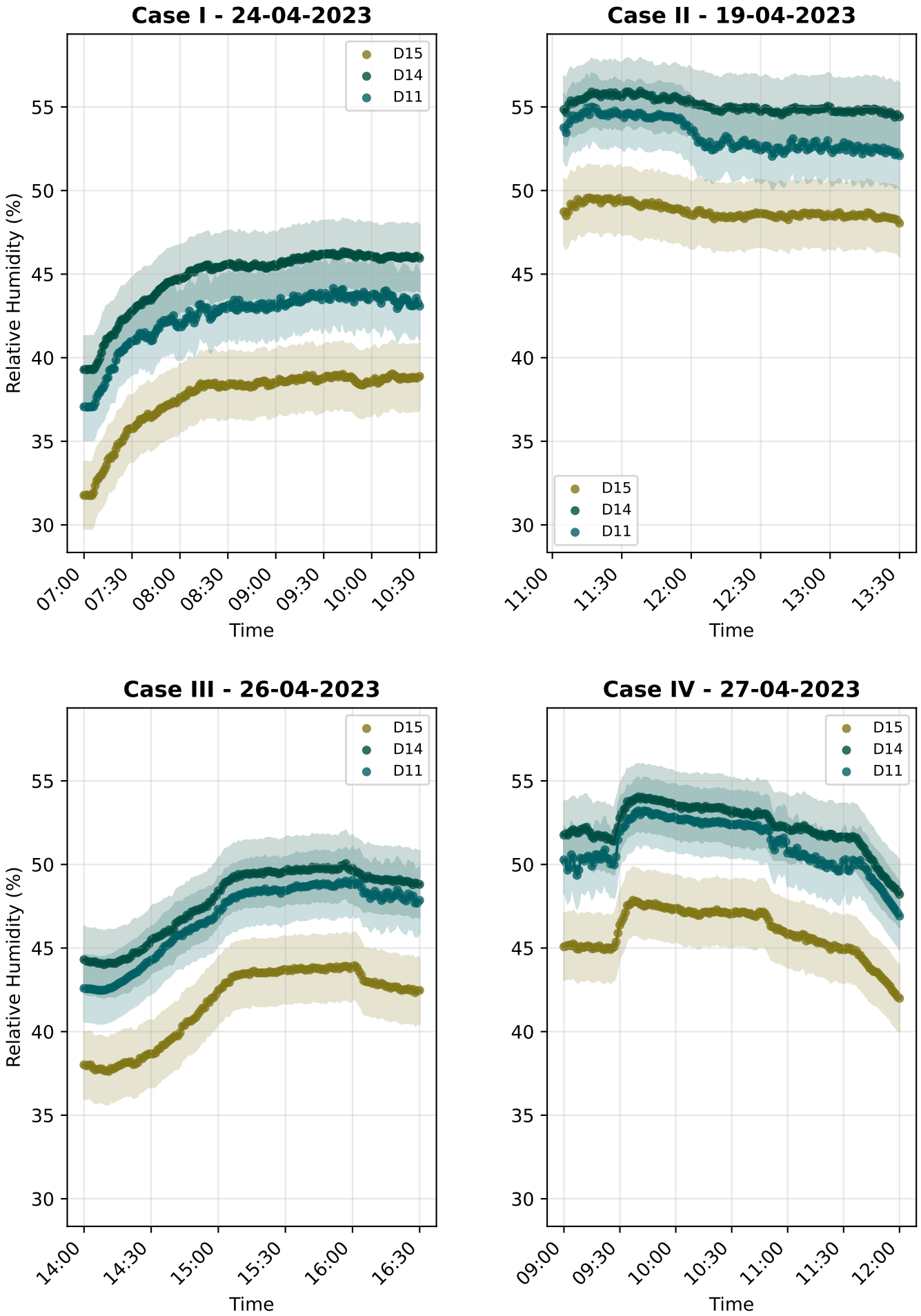


Relative Humidity (%) Time Series
Sensors 15, 14, and 11

elative Humidity (%) - 30-Second Resolution - Sensors 15, 14, and

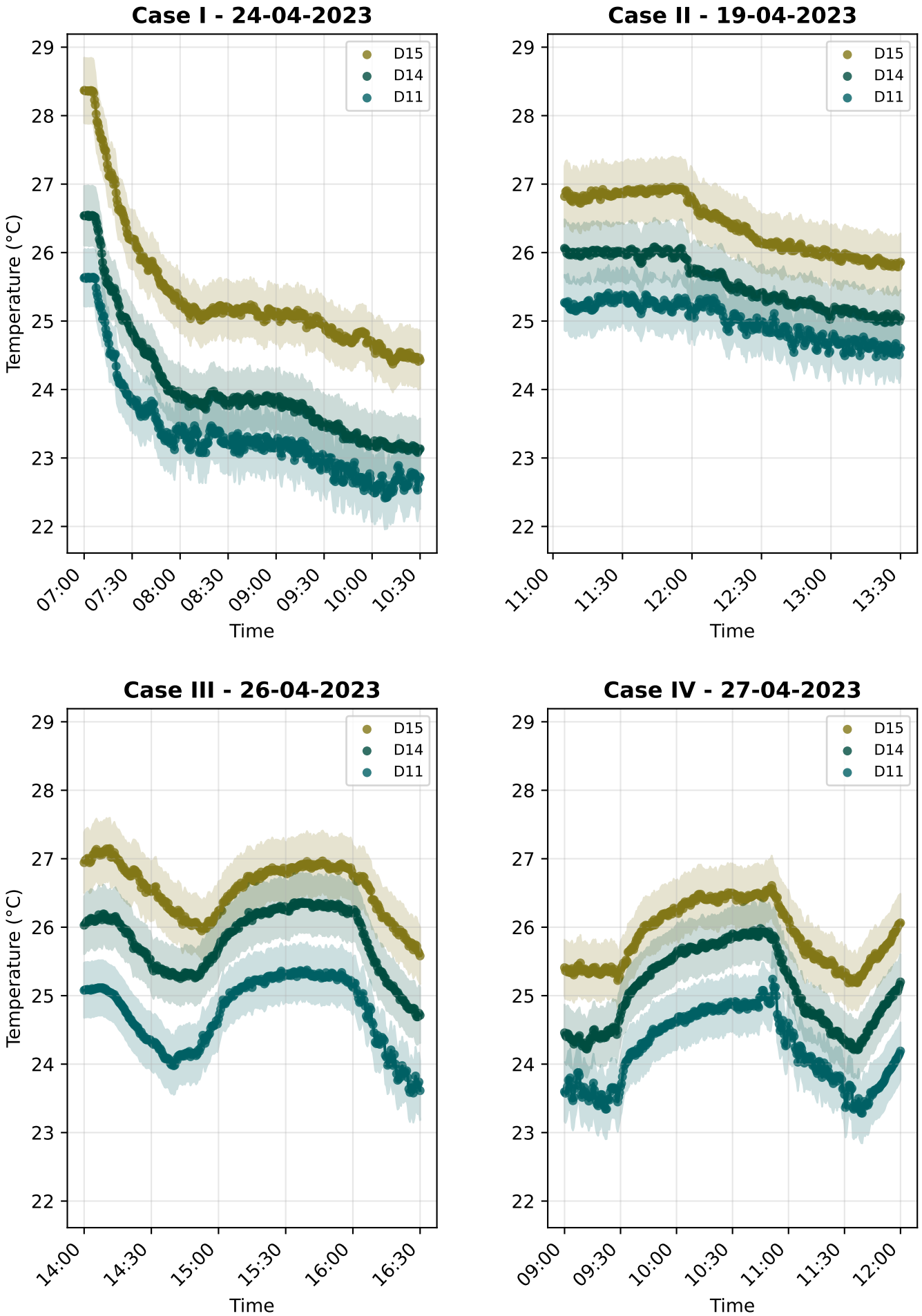


elative Humidity (%) - 1-Minute Resolution - Sensors 15, 14, and 1

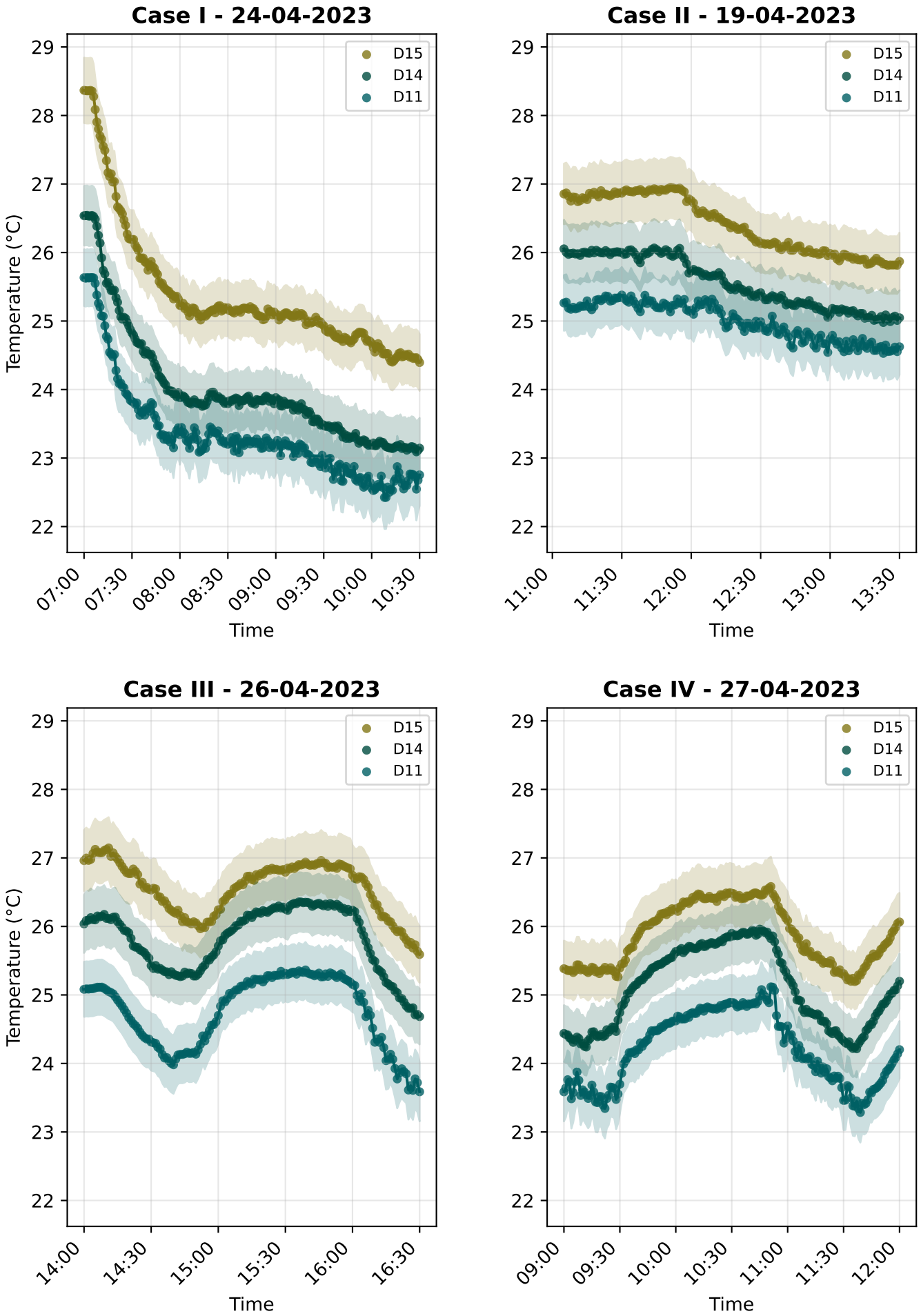


Temperature (°C) Time Series
Sensors 15, 14, and 11

Temperature (°C) - 30-Second Resolution - Sensors 15, 14, and 11

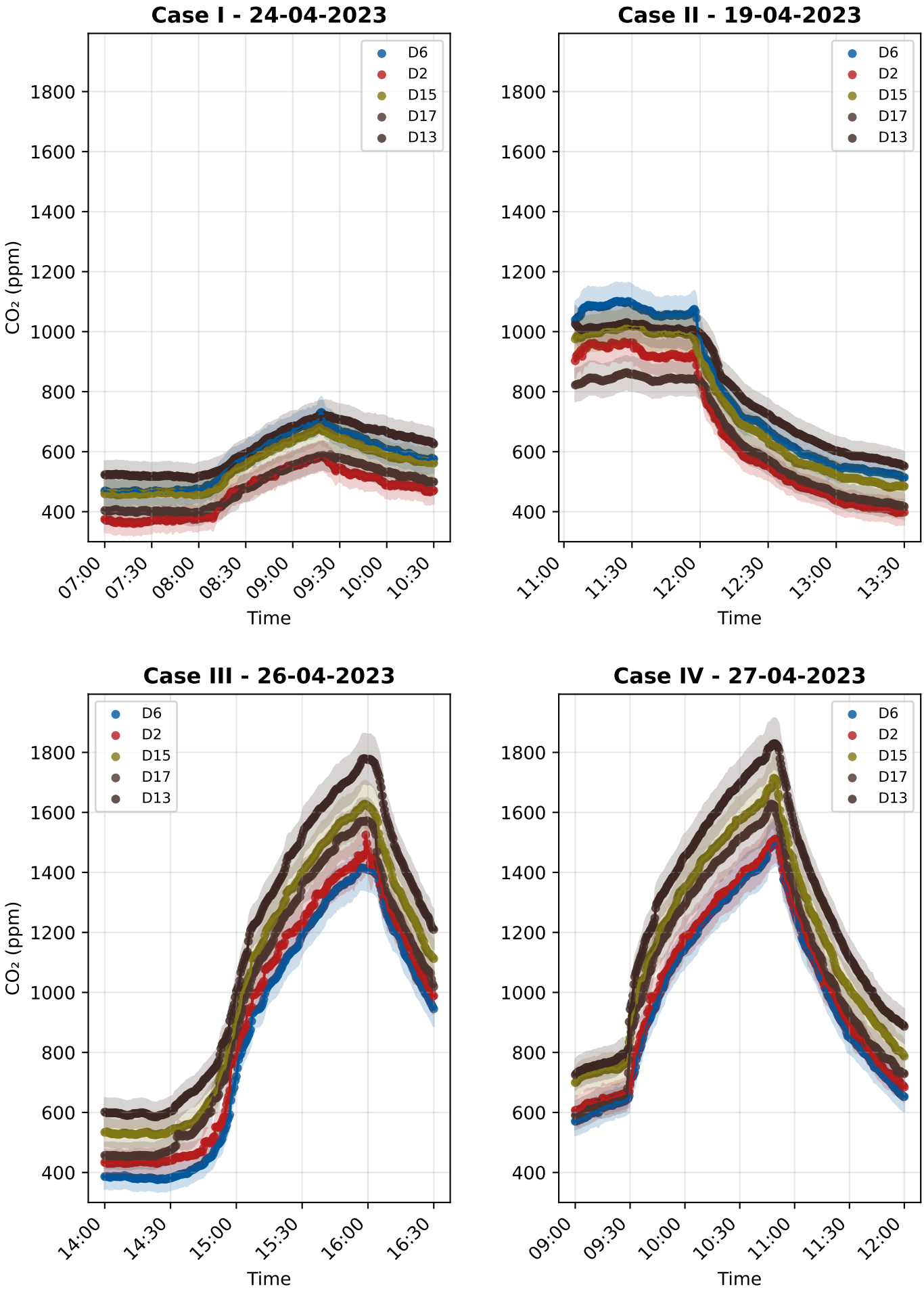


Temperature (°C) - 1-Minute Resolution - Sensors 15, 14, and 11

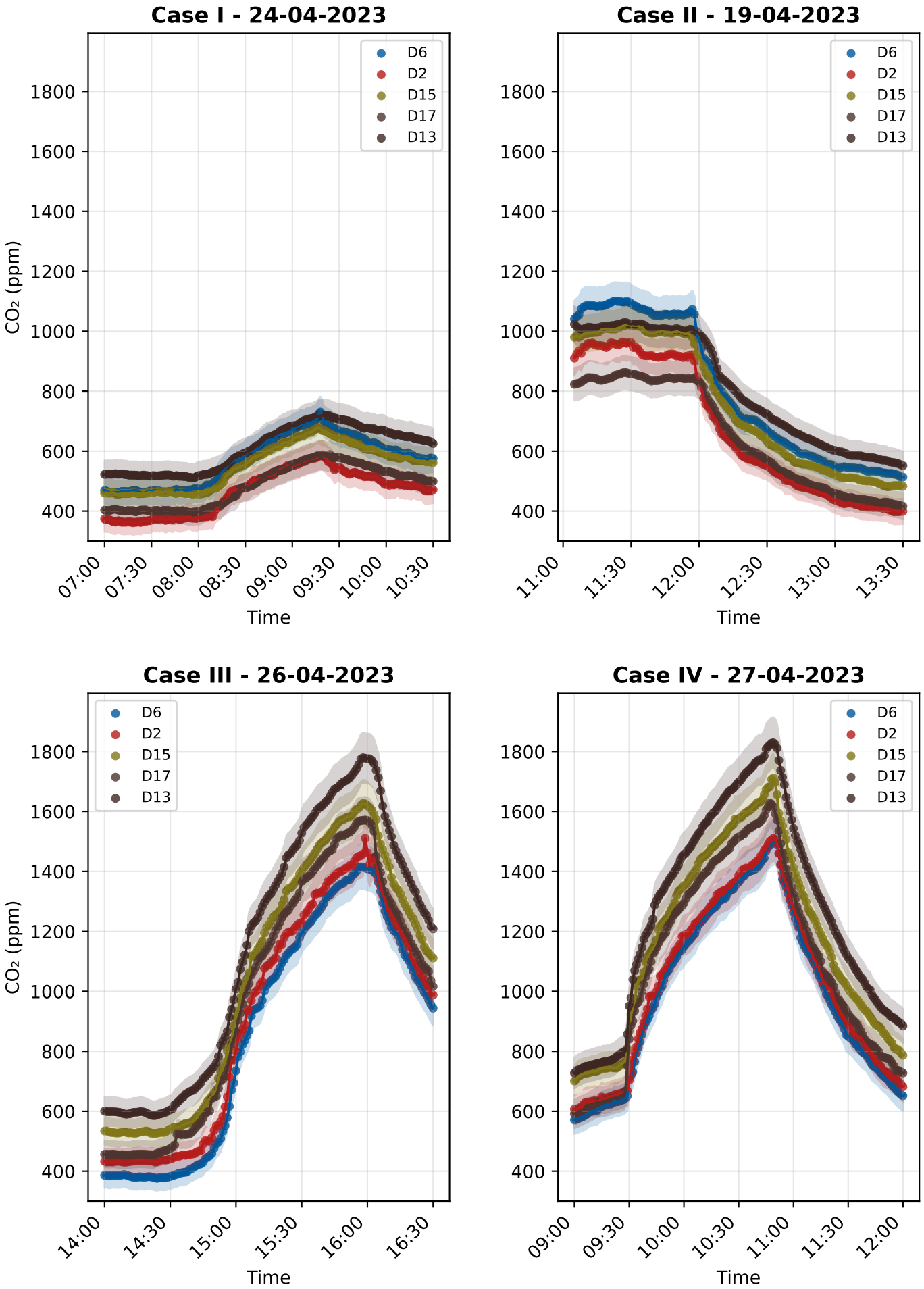


CO_2 (ppm) Time Series
Sensors 6, 2, 15, 17, and 13

CO_2 (ppm) - 30-Second Resolution - Sensors 6, 2, 15, 17, and 13

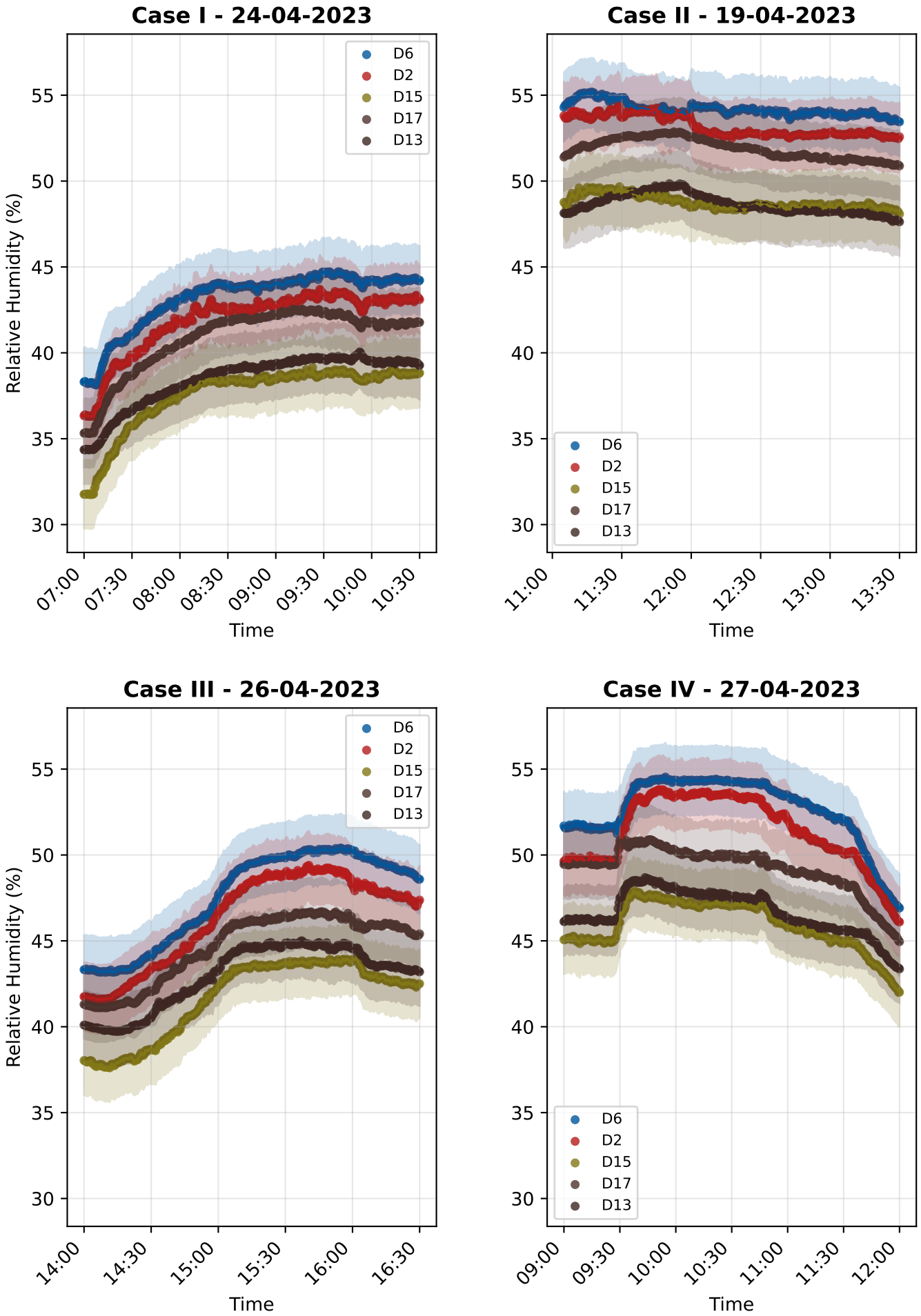


CO_2 (ppm) - 1-Minute Resolution - Sensors 6, 2, 15, 17, and 13

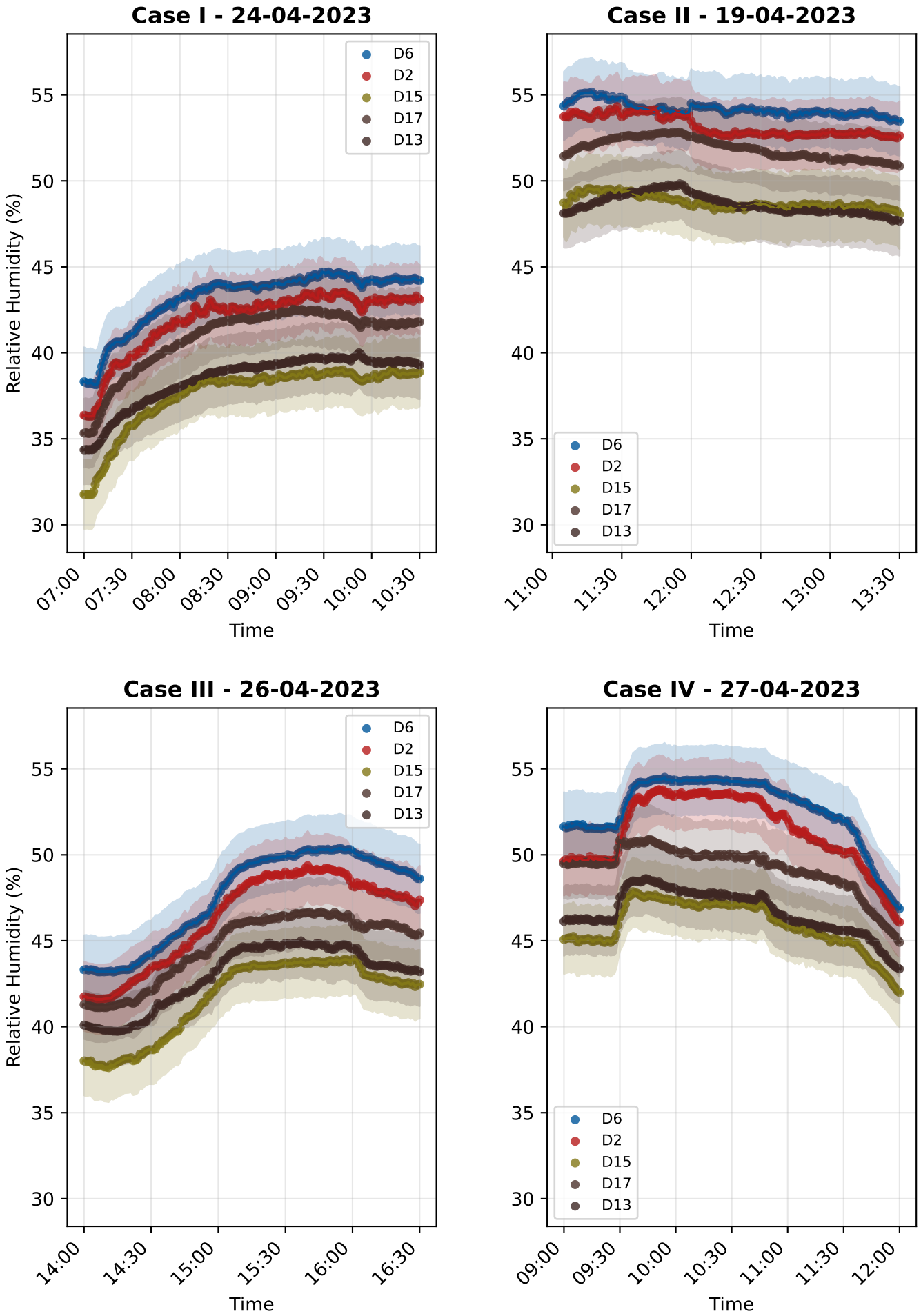


Relative Humidity (%) Time Series
Sensors 6, 2, 15, 17, and 13

tive Humidity (%) - 30-Second Resolution - Sensors 6, 2, 15, 17, an

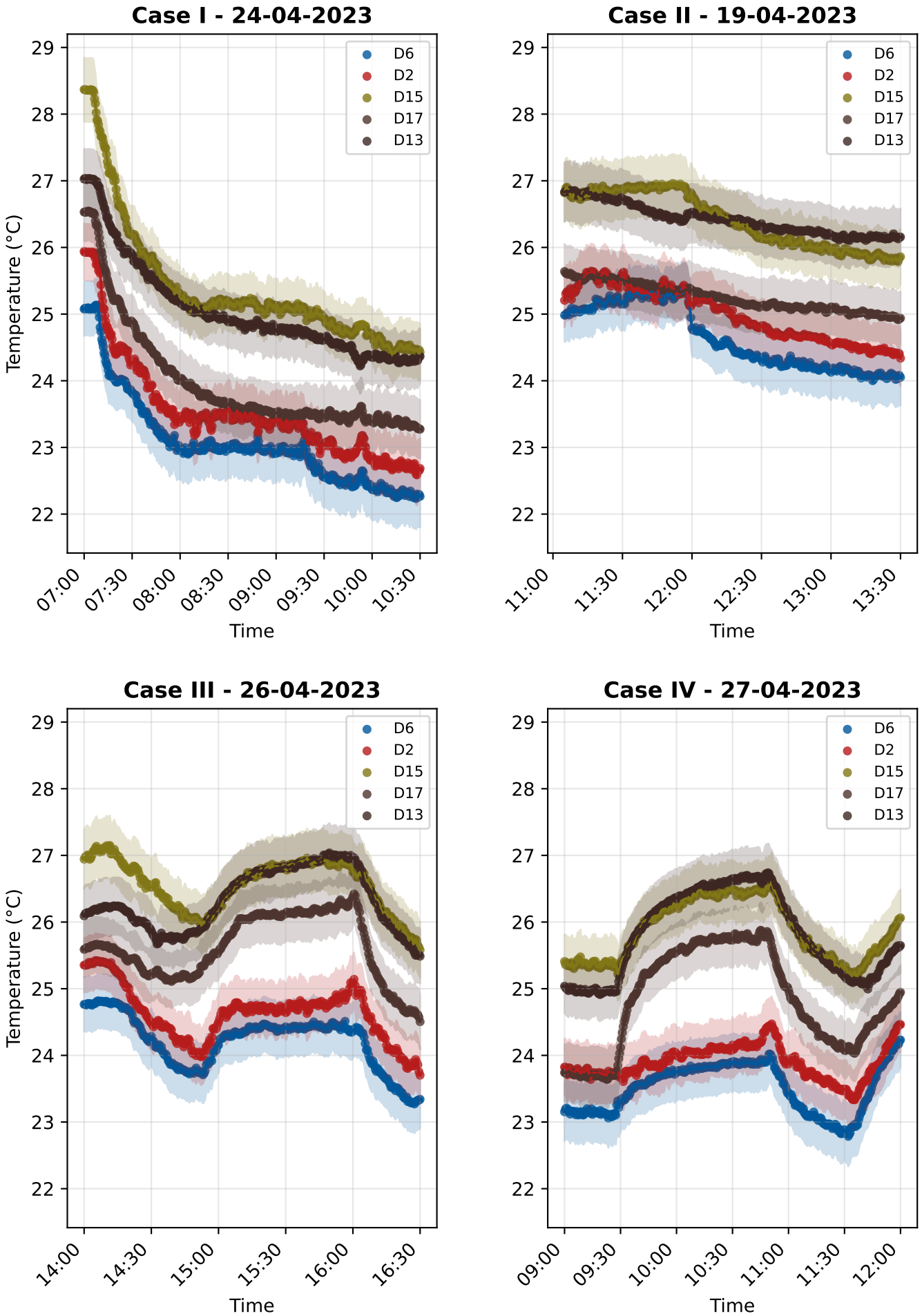


tive Humidity (%) - 1-Minute Resolution - Sensors 6, 2, 15, 17, and

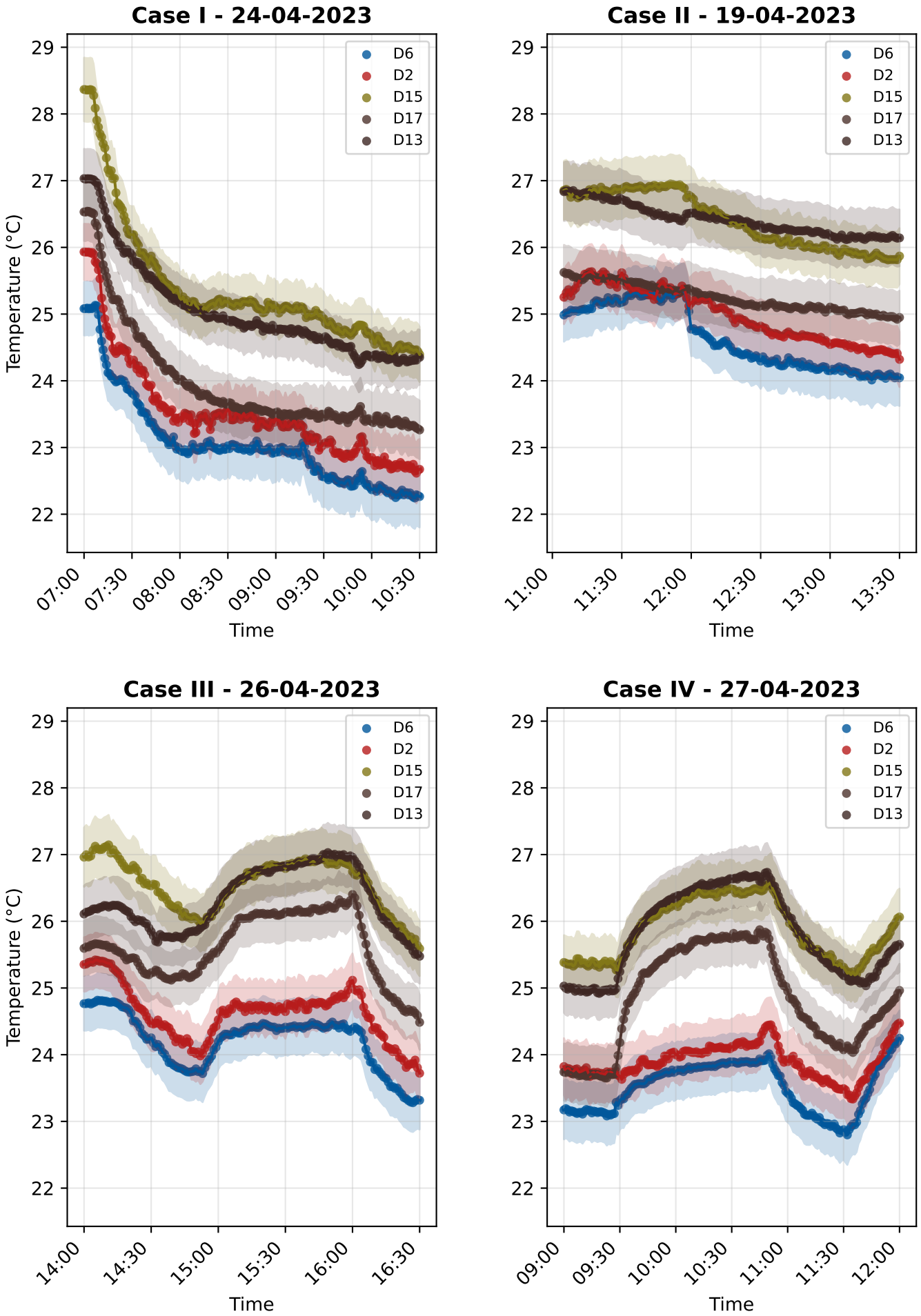


Temperature (°C) Time Series
Sensors 6, 2, 15, 17, and 13

mperature (°C) - 30-Second Resolution - Sensors 6, 2, 15, 17, and

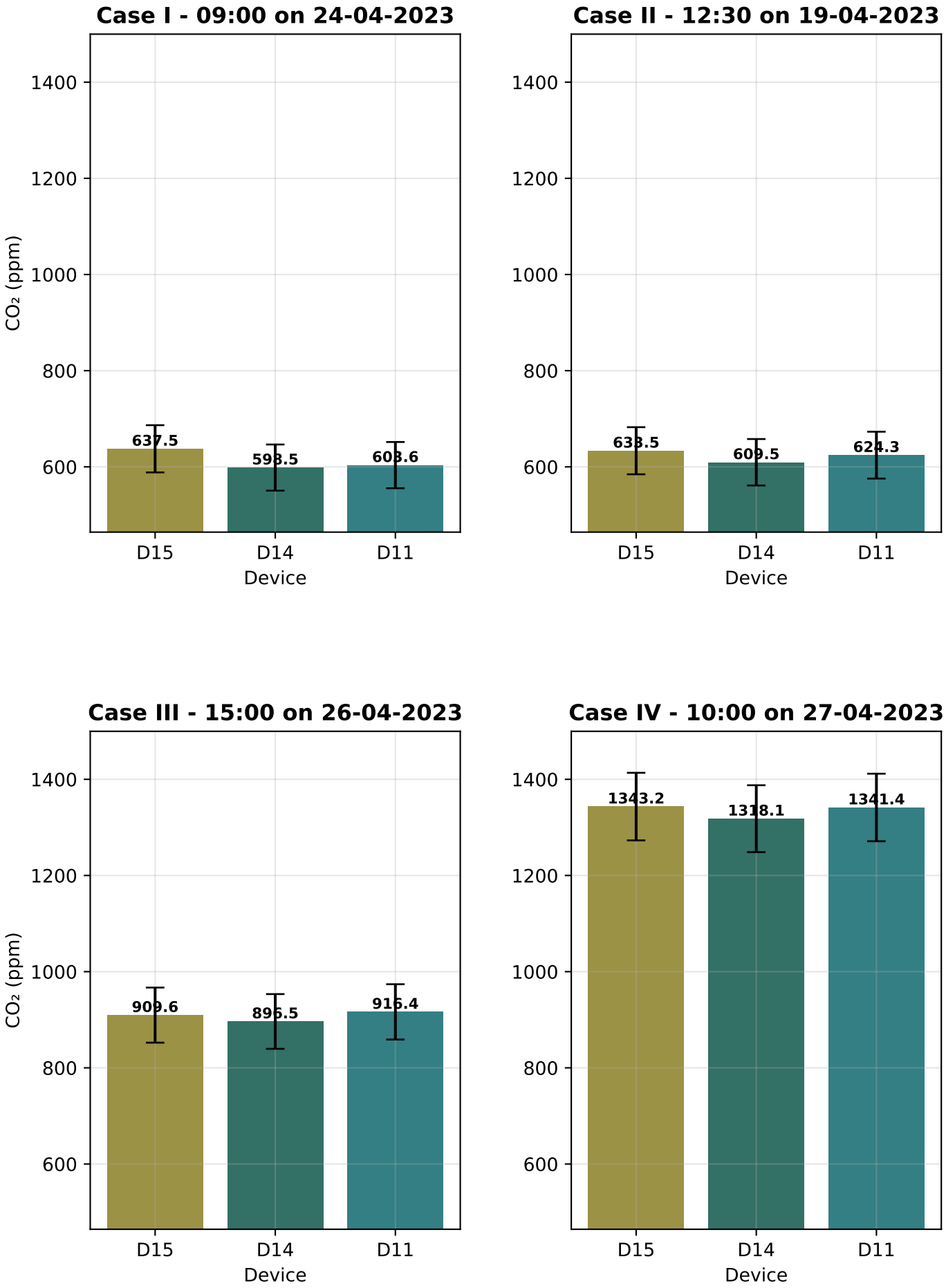


emperature (°C) - 1-Minute Resolution - Sensors 6, 2, 15, 17, and 1

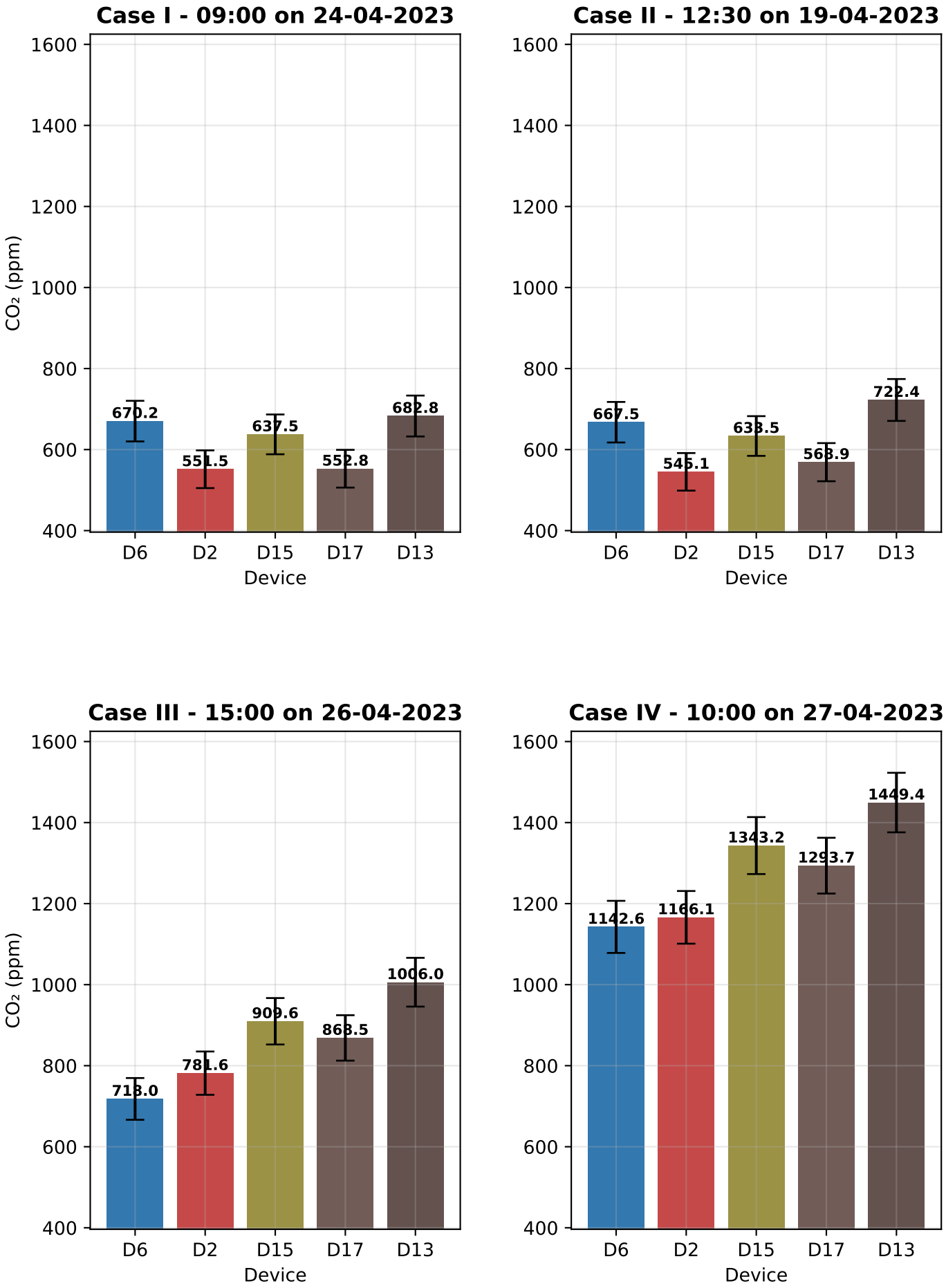


CO_2 (ppm) Bar Plots with Uncertainty

CO_2 (ppm) Levels - Sensors 15, 14, and 11

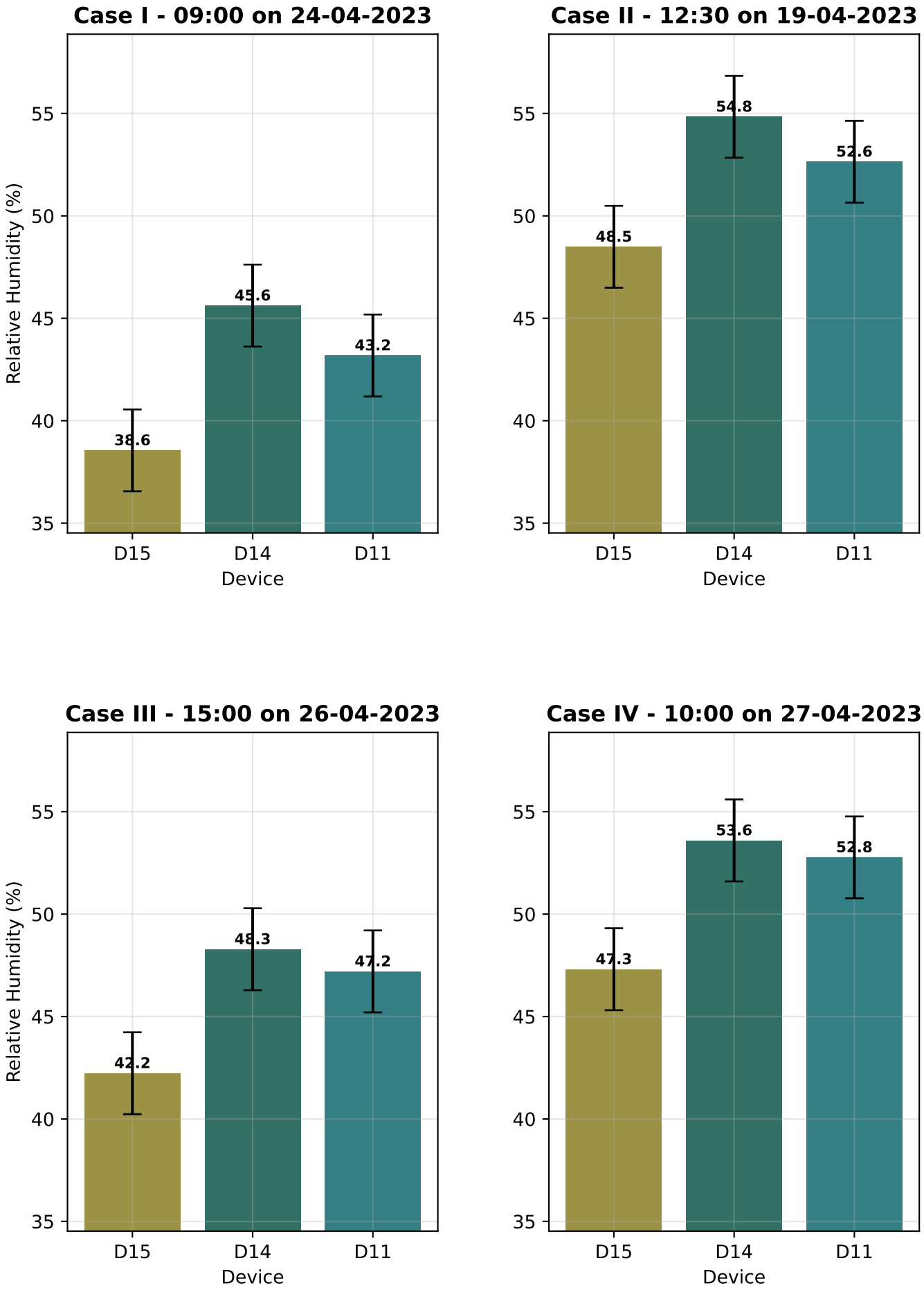


CO_2 (ppm) Levels - Sensors 6, 2, 15, 17, and 13

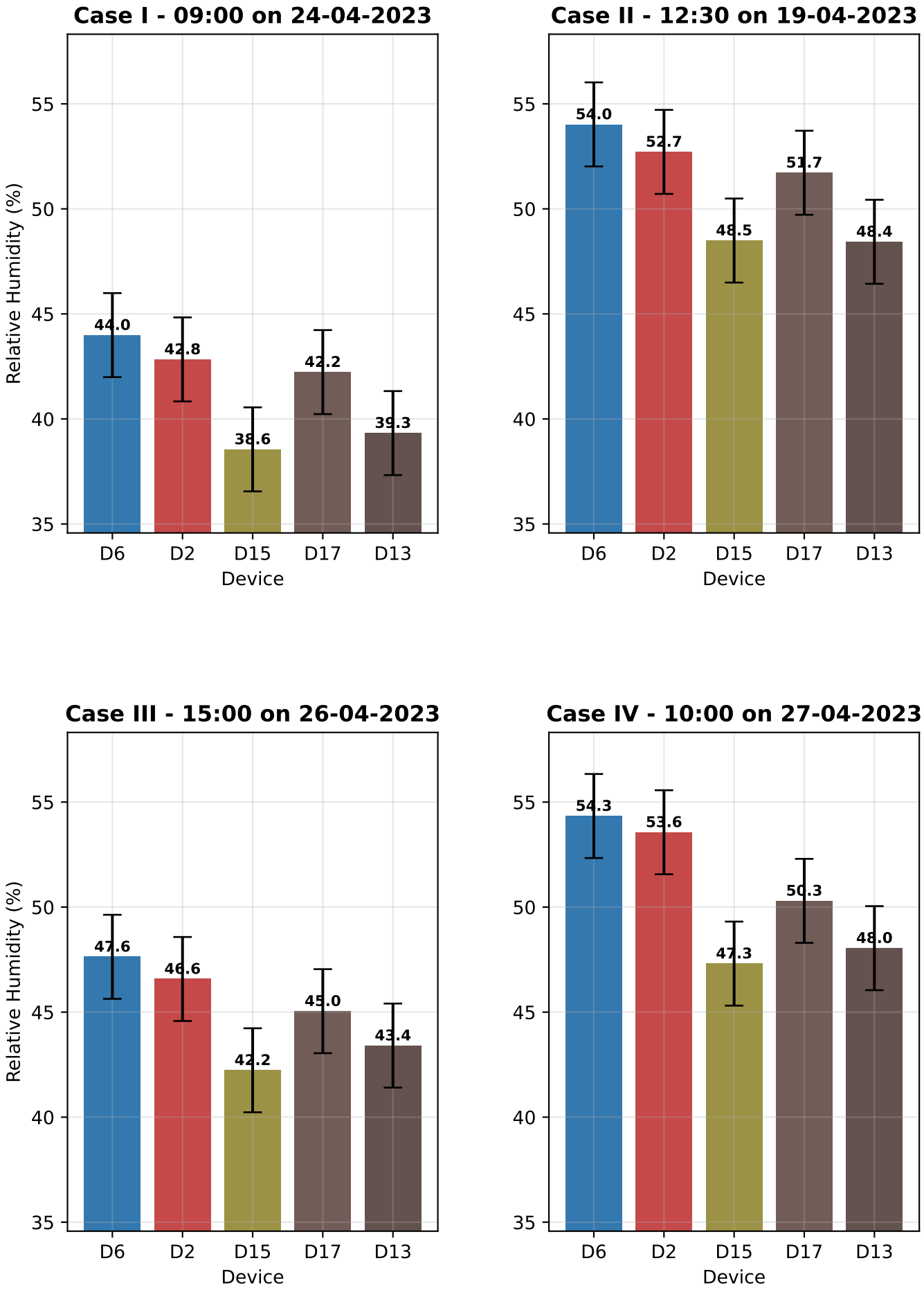


Relative Humidity (%) Bar Plots with Uncertainty

Relative Humidity (%) Levels - Sensors 15, 14, and 11

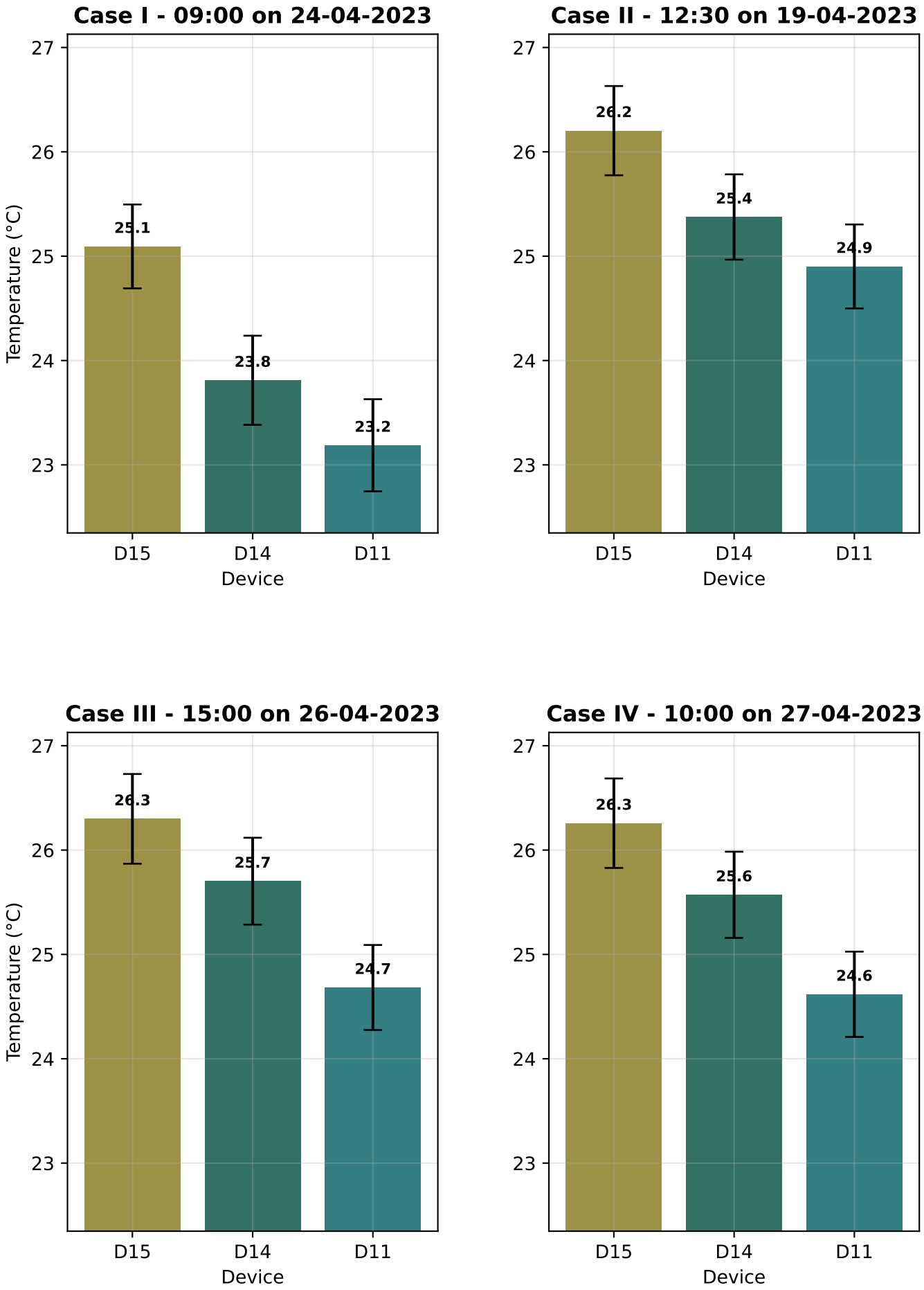


Relative Humidity (%) Levels - Sensors 6, 2, 15, 17, and 13

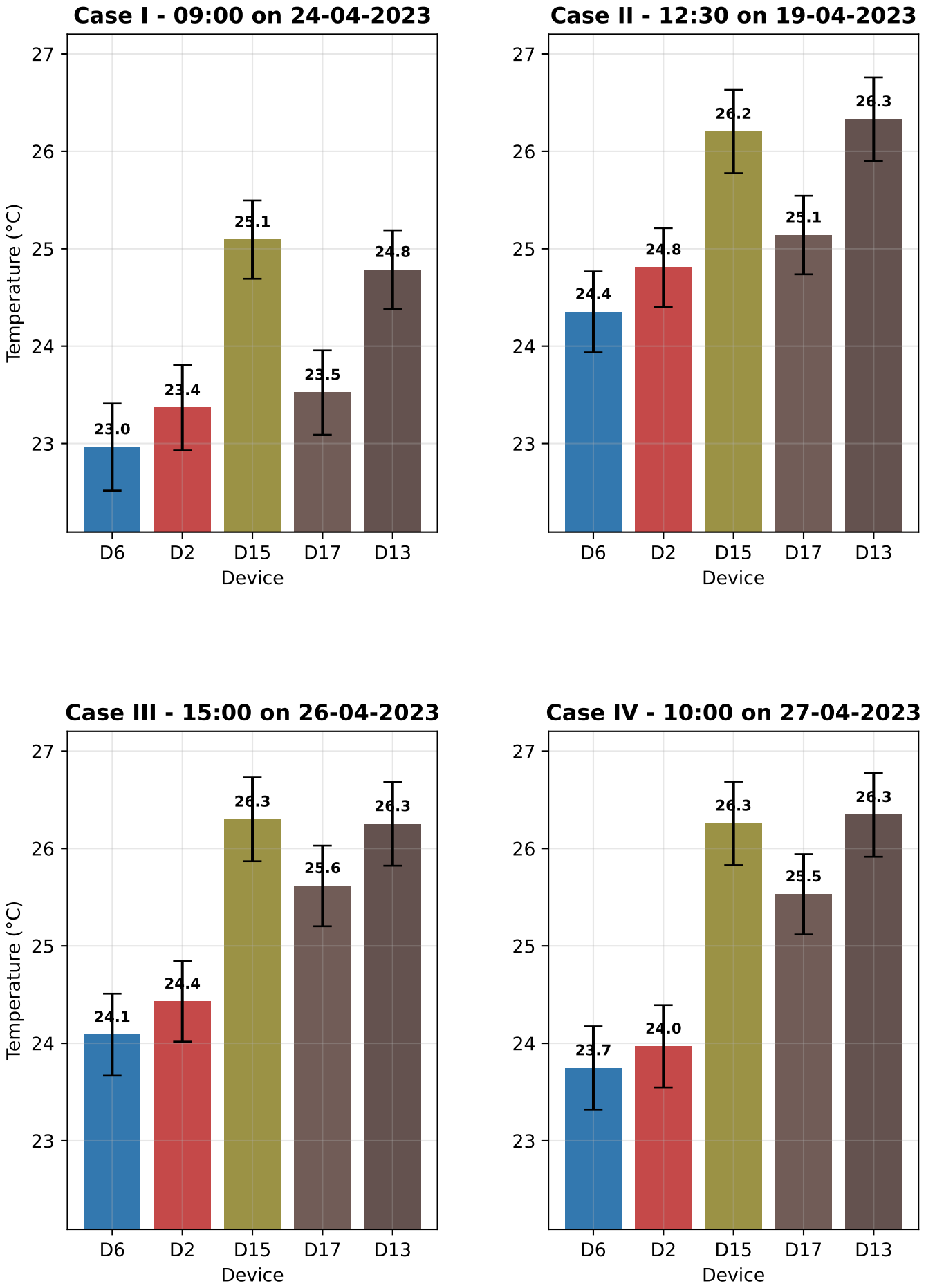


Temperature (°C) Bar Plots with Uncertainty

Temperature (°C) Levels - Sensors 15, 14, and 11



Temperature (°C) Levels - Sensors 6, 2, 15, 17, and 13



REFERENCES

1. 1970s energy crisis - Wikiwand. https://www.wikiwand.com/en/1970s%20energy%20crisis
2. Weimer, D. L. (2013). Strategic Petroleum Reserves. Elsevier EBooks. https://doi.org/10.1016/b978-0-12-409548-9.01391-9
3. Candles FAQ | FAQ - Homecourt. https://homecourt.co/en-gb/blogs/candles/candles-faq
4. Common Mold Types - Identifying Prevalent Varieties. https://chooseeliable.com/common-mold-types-identifying-prevalent-varieties/
5. Abbas, M., Javed, R., Javed, Z., Mughal, T., & Noor, F. (2019). Air Quality Monitoring of PM2.5 and PM10 at Heavy Traffic sites of Asian's Oldest Market: Anarkali Bazaar, Lahore. https://doi.org/10.46660/ijeeg.Vol0.Iss0.0.72
6. Radon Testing Vermont | Radon Testing and Mitigation VT. https://www.nationalradondefense.com/radon-contractors/vermont.html
7. Carcinogen Definition: What Are Carcinogens and Their Role in Cancer Development - Healthtruly.com. https://healthtruly.com/carcinogen-definition-what-are-carcinogens-and-their-role-in-cancer-development.html
8. Asbestos Testing & Inspection | Certified Firm - New Jersey. https://www.aaaleadpro.com/service/asbestos-testing/
9. Professional Air Duct Cleaning Service in Davie FL. https://www.best-duct-cleaning.com/everything-you-need-to-know-about-cleaning-air-ducts
10. HRCare - OSHA Updates Its Guidance on Workplace Violence. https://www.hrcare.com/article.aspx/205/3899/
11. Protecting Lives with Reliable CO_2Gas Detectors - Gas Detector&sensor/Soil Sensor/Water Quality/Weather Station - JXCT. http://jxdziot.com/blog/9215.html
12. Time-invariant : Data Analysis Explained - EasyBA.co. https://easyba.co/blog/data-analysis/data-analysis-glossary/time-invariant-data-analysis-explained/
13. Combating COVID-19 and Airborne Pathogens: The Power of Air Filtration, Portable HEPA Filters, and Ventilation - Filters Unlimited LLC. https://filtersunlimitedde.com/combating-covid-19-and-airborne-pathogens-the-power-of-air-filtration-portable-hepa-filters-and-ventilation/
14. Scarification Methods for Wildflower or Native Plant Seeds - PlantPropagation.org. https://plantpropagation.org/scarification-methods-for-wildflower-or-native-plant-seeds/
15. Marine Geochemistry: Mastering the Elements. https://davidhastingsmarinescience.com/mastering-the-elements-of-marine-geochemistry/
16. Thoughts on how the AI phenomenon will impact on your Air Conditioning? - Air Conditioning Noosa. https://www.noosa-air-conditioning.com.au/article/thoughts-on-how-the-ai-phenomenon-will-impact-on-your-air-conditioning
17. (2015). United States : Marsh Named Best Executive Risk Retail Broking Team by Advisen. MENA Report.
18. Indoor Air Quality In Roseville, CA | Economy Heating & Air. https://economyheatandair.com/roseville-indoor-air-quality-services/
19. Protecting Lives with Reliable CO_2Gas Detectors - Gas Detector&sensor/Soil Sensor/Water Quality/Weather Station - JXCT. http://jxdziot.com/blog/9215.html

20. Protecting Lives with Reliable CO_2Gas Detectors - Gas Detector&sensor/Soil Sensor/Water Quality/Weather Station - JXCT. http://jxdziot.com/blog/9215.html